Memo - Trigonometry Revision

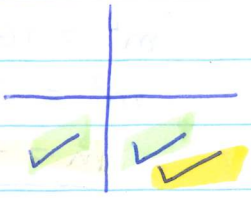
5.4.1 $T = \dfrac{2\sin(35{,}8° + 20°)}{\cos 60° - \tan 40°} = -7{,}317$

5.4.2 $T = \dfrac{2\sin(130° + 20)}{\cos 150° - \tan 40°} = -0{,}880$

6.2 $25\sin A + 7 = 0$ $\cos A > 0$

$25\sin A = -7$

$\sin A = \dfrac{-7}{25}$ $\dfrac{y}{r}$



$\therefore$ Quad (4)

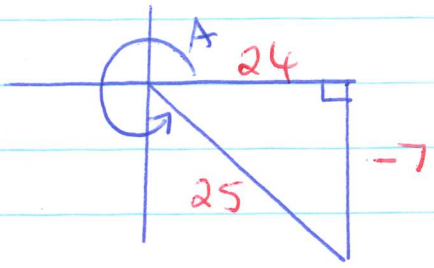
$x^2 + y^2 = r^2$ (Pythag)

$x^2 + (-7)^2 = 25^2$

$x^2 = 576$

$x = \pm 24$

$x = 24$ (quad 4)



6.21 $\cos A = \dfrac{24}{25}$ $\dfrac{x}{r}$

6.22. $\sin^2 A + (\cos A)^2$

$= \left(\dfrac{-7}{25}\right)^2 + \left(\dfrac{24}{25}\right)^2$

$= \dfrac{49}{625} + \dfrac{576}{625}$

$= 1$

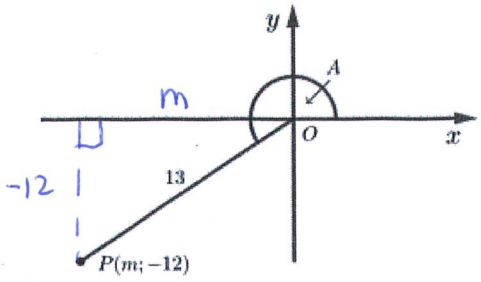
6.24 $1 + \tan A - \dfrac{1}{\sin A}$

$= 1 + \dfrac{-7}{24} - \dfrac{1}{\frac{-7}{25}}$

$= 1 - \dfrac{7}{24} + \dfrac{25}{7} = \dfrac{719}{168}$

Pg (2)

6.3 In the diagram, $OP = 13$ with $P(m\,;-12)$ and $x\hat{O}P = A$.



Without using a calculator, determine the value of:

6.3.1 m (2)

$m^2 + (-12)^2 = 13^2$ Pythag

$m^2 = 169 - 144$

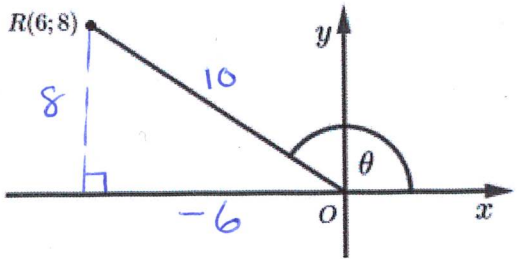
$m^2 = 25$

$m = -5$ (Quad 3)

6.3.2 $\cos A$ (2)

$\cos A = \frac{-5}{13}$

6.4 In the diagram $R(6\,;8)$ lies on the line OR. $x\hat{O}R = \theta$.



Without using a calculator, determine the value of:

6.4.1 the length OR. (2)

6.4.2 $1 - 2\sin\theta \times \cos\theta$ (3)

6.4.3 $\cos 2\theta$ if $\cos 2\theta = \cos^2\theta - \sin^2\theta$ (3)

6.4.1 $(-6)^2 + 8^2 = r^2$ Pythag

$36 + 64 = r^2$

$100 = r^2$

$10 = r$

6.4.2 $1 - 2\sin\theta\cos\theta$

$= 1 - 2\left(\frac{8}{10}\right)\left(\frac{-6}{10}\right)$

$= 1 - 2\left(\frac{-48}{100}\right)$

$= \frac{49}{25}$

6.4.3

$\cos^2\theta - \sin^2\theta$

$= \left(\frac{-6}{10}\right)^2 - \left(\frac{8}{10}\right)^2$

$= \frac{36}{100} - \frac{64}{100}$

$= \frac{-28}{100}$

$= -\frac{7}{25}$

$= \cos 2\theta$

6.6 If $-tanA = \frac{4}{3}$ and $0^\circ \leq A \leq 180^\circ$, without the use of a calculator, determine the value of the following:

6.6.1 $sinA$ (3)

$-tanA = \frac{4}{3}$

$\therefore tan = -\frac{4}{3}$ $\frac{y}{x}$ $\therefore$ Quad (2)

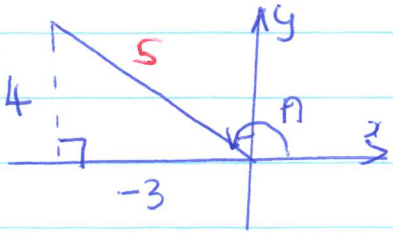
$\sin A = \frac{4}{5}$

$3^2 + (-4)^2 = r^2$

$9 + 16 = r^2$

$25 = r^2$

$5 = r$ Pythag



6.7 Given that $B \in [0^\circ; 360^\circ]$, which quadrant(s) will/can angle B lie in if:

6.7.1 $sinB > 0$ and $cosB < 0$? Quad 2 (2)

6.7.2 $cosB > 0$ and $tanB < 0$? Quad 4 (2)

6.7.3 $sinB \times tanB > 0$? Quad 1 & Quad 4 (3)

+ x + Both +

or − x − Both −

7.1 Solve the following equations for $x \in [0^\circ; 90^\circ]$:

7.1.1 $-3.sin2x + 1 = 0$ (2)

7.1.2 $tan\left(\frac{1}{2}x + 40^\circ\right) = 4$ (2)

7.1.3 $\sqrt{5}\,cos(x - 5^\circ) = 2$ (2)

7.1.4 $4.sin\frac{x}{3} = \sqrt{3}$ (3)

7.1.1 $-3\sin 2x + 1 = 0$

$-3\sin 2x = -1$

$\sin 2x = \frac{1}{3}$

$KA = 19,47^\circ$

$\therefore 2x = 19,47^\circ$

$x = 9,74^\circ$

7.1.2 $\tan(\frac{1}{2}x + 40) = 4$

$KA = 75,96^\circ$

$\frac{1}{2}x + 40^\circ = 75,96^\circ$

$\frac{1}{2}x = 35,96^\circ$

$x = 71,93^\circ$

7.1.3 $\sqrt{5}\cos(x - 5^\circ) = 2$

$\cos(x - 5^\circ) = \frac{2}{\sqrt{5}}$

$KA = 26,57^\circ$

$x - 5^\circ = 26,57^\circ$

$x = 31,57^\circ$

7.1.4 $4\sin(\frac{x}{3}) = \sqrt{3}$

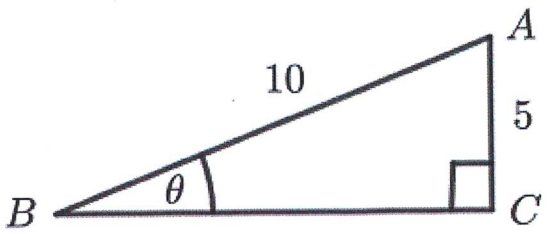
$\sin(\frac{x}{3}) = \frac{\sqrt{3}}{4}$

$KA = 25,66^\circ$

$\therefore \frac{x}{3} = 25,66^\circ$

$x = 76,98^\circ$

8.1 Right-angled triangle ABC is shown in the figure.



8.1.1 determine the value of θ. (2)

8.1.2 hence use θ to calculate the length BC in simplest surd form. (3)

8.1.3 determine BC by applying Pythagoras's Theorem in simplest surd form. (2)

8.1.1 $\sin \hat{B} = \frac{5}{10}$

$\hat{B} = 30°$

8.1.2 $\cos 30° = \frac{BC}{AB} = \frac{BC}{10}$

$10 \cos 30° = BC$

$5\sqrt{3} = BC$

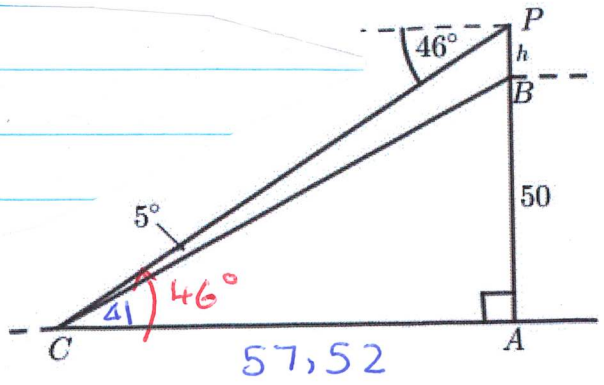
8.1.3 $BC^2 + AC^2 = AB^2$ Pythag

$BC^2 \quad 5^2 = 10^2$

$BC^2 = 75$

$BC = 5\sqrt{3}$

8.2 In the sketch, CA is the horizontal ground level. The side AB of a building is perpendicular to CA with $AB = 50$ metres. PB is a vertical pole height h and $B\hat{C}P = 5^\circ$. The angle of depression of C from P is 46°.



Determine the value of h. (6)

$P\hat{C}A = 46^\circ$ alt ∠s : // lines

$B\hat{C}A = P\hat{C}A - 5^\circ$

$B\hat{C}A = 41^\circ$

In ΔBCA : $\tan 41^\circ = \frac{BA}{CA}$

$\tan 41^\circ = \frac{50}{CA}$

$CA = \frac{50}{\tan 41^\circ}$

$CA = 57,52\ m$

In ΔPCA $\tan 46^\circ = \frac{PA}{CA}$

$57,52 \tan 46^\circ = PA$

$59,56\ m = PA$

$h = PA - BA$

$= 59,56 - 50$

$= 9,56\ m.$

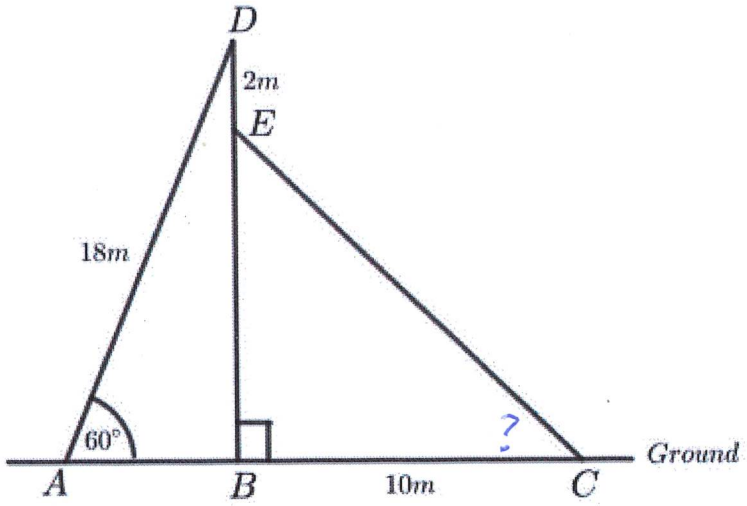
Pg (6)

8.4 In the diagram, A, B, C, D, and E all lie in the same vertical plane. ABC is the horizontal ground level. BD is a vertical flagpole which is supported by two steel cables, AD and CE. The angle of elevation of D from A is 60°.
$AD = 18$ *metres*; $BC = 10$ *metres*; $DE = 2$ *metres*

8.4.1 Calculate the height of the flagpole BD, leaving your answer in simplest surd form) (2)

8.4.2 Hence calculate the length of the steel cable CE. (4)

8.4.3 Determine the angle of elevation of E from C, correct to ONE decimal place. (2)



8.4.1 In ΔABD

$\sin 60^\circ = \frac{DB}{AD}$

$\sin 60^\circ = \frac{DB}{18}$

$18 \sin 60^\circ = DB$

$9\sqrt{3} = DB$

8.4.2 $BE = DB - 2$

$= 9\sqrt{3} - 2$

$= 7\sqrt{3}$

In ΔBEC

$EC^2 = BE^2 + BC^2$ Pythag

$= (7\sqrt{3})^2 + 10^2$

$= 247$

$EC = 15,72$ m

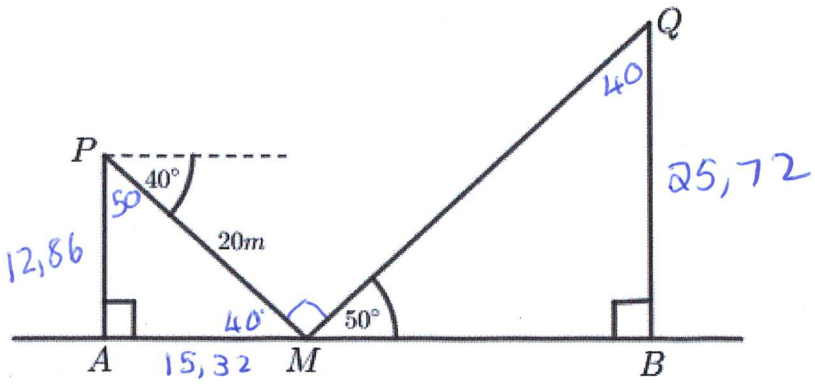
8.4.3

$\tan \hat{C} = \frac{BE}{BC}$

$\tan \hat{C} = \frac{7\sqrt{3}}{10}$

$\hat{C} = 50,5^\circ$

8.7 In the figure AMB is the horizontal ground level. AP and BQ are vertical poles with $AP = \frac{1}{2}BQ$. $PM = 20$ metres.
The angle of depression of M from P is 40° and the angle of elevation of Q from M is 50°.



Determine the distance AB. (7)

In Δ PAM : $P\hat{M}A = 40^\circ$ alt $\angle$s : // lines

$\sin 40^\circ = \frac{PA}{PM}$ OR $\cos 40^\circ = \frac{AM}{PM}$

$\sin 40^\circ = \frac{PA}{20}$ $\cos 40^\circ = \frac{AM}{20}$

$20 \sin 40 = PA$ $20 \cos 40^\circ = AM$

$12,86 = PA$ $15,32m = AM$

In Δ MQB $\tan 50^\circ = \frac{QB}{MB}$

$QB = 2(12,86)$ $\tan 50 = \frac{25,72}{MB}$

$QB = 25,72\ m$ $MB \tan 50^\circ = 25,72$

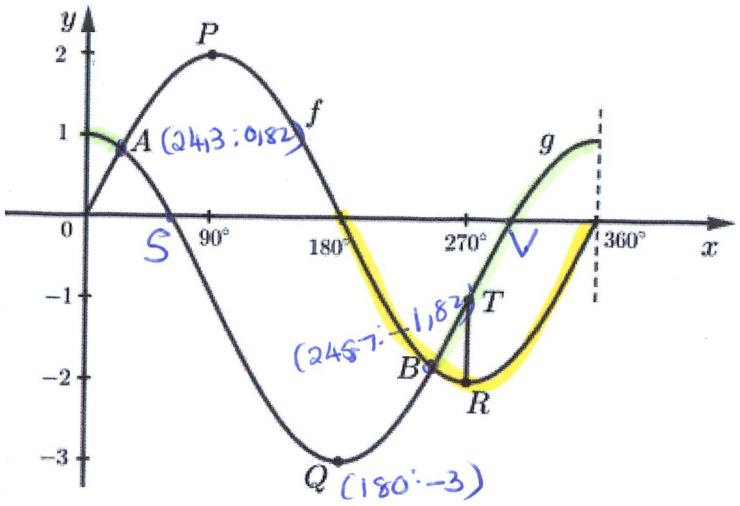
Given $AP = \frac{1}{2} BQ$ $MB = \frac{25,72}{\tan 50}$

$AB = AM + MB$ $MB = 21,58m$

$= 15,32 + 21,58$

$= 36,90\ m$

9.4 In the diagram the graphs of $f(x) = m.\sin x$ and $g(x) = 2.\cos x + n$ are shown for $x \in [0^\circ; 360^\circ]$. $Q(180^\circ; -3)$ is a turning point on g. P and R are turning points on f. The two graphs intersect at $A(24{,}30^\circ; 0{,}82)$ and $B(245{,}70^\circ; -1{,}82)$.



9.4.1 Write down the values of m and n. $m = 2$; $n = -1$ (2)

9.4.2 Write down the coordinates of P and Q. $P(90^\circ; 2)$ $Q(180^\circ; -3)$ (2)

9.4.3 $TR \parallel y$-axis with T on g. Determine the length TR. (2)

9.4.4 Determine the coordinates where $g(x) = 0$. (3)

9.4.5 For which values of x is:

9.4.5.1 $f(x) \leq 0$? (2) 9.4.5.2 $g(x) > f(x)$? (2)

9.4.3 $T(270^\circ; -1)$ & $R(270^\circ; -2)$

$\therefore TR = 1$ unit Vertical length.

9.4.4 $g(x) = 0$

$2\cos x - 1 = 0$

$2\cos x = 1$

$\cos x = 1/2$

$KA = 60^\circ$

$\therefore S(60^\circ; 0)$ & $V(300^\circ; 0)$

9.4.5(1) $f(x) \leq 0$ $x \in [180^\circ; 360^\circ)$ See yellow line

(2) $g(x) > f(x)$ graph g is above graph f

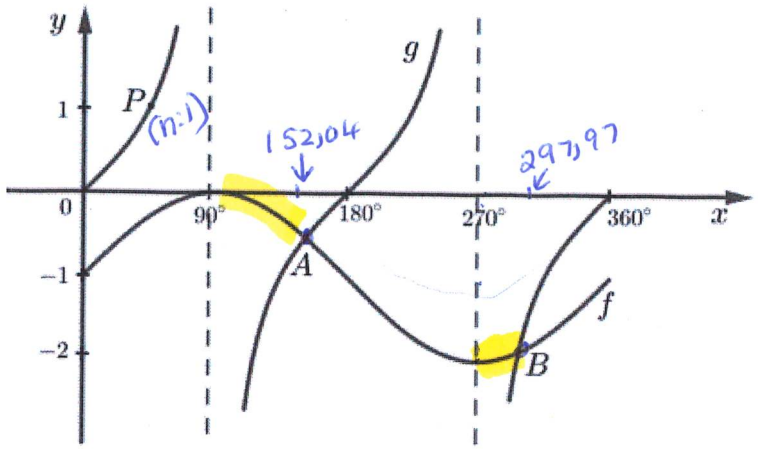
$x \in (0; 24{,}3^\circ)$ &

$x \in (245{,}7^\circ; 360^\circ)$

See green lines

pg (9)

9.5 In the sketch, $f(x) = sinx + m$ and $g(x) = tanx$ are shown for $x \in [0° ; 360°]$. $P(n ; 1)$ lies on g. A and B are the points of intersection of the two graphs with $x_A = 152{,}04°$ and $x_B = 297{,}97°$.



9.5.1 Write down the value of m. $m = -1$ (1)

9.5.2 What is the range of $f(x)$? $y \in [-2 ; 0]$ (1)

9.5.3 Write down the value of n. $\tan 45° = 1 \therefore n = 45°$ (1)

9.5.4 For which values of x is:

9.5.4.1 $f(x) > g(x)$? (4)

9.5.4.2 $f(x).g(x) = 0$? (3)

9.5.5 $h(x) = f(x) + k$. For which value(s) of k will h have exactly one x-intercept? (2)

9.5.4.1 $f(x) > g(x)$ where $x \in (90° ; 152{,}04°)$
See yellow lines & $x \in (270° ; 297{,}97°)$

9.5.5 When $h(x) = f(x) + 0$
$= \sin x - 1$ (the graph as it is with x-intercept of 90°)

ie $k = 0$

AND

When $h(x) = f(x) + 2$
$= \sin x - 1 + 2$
$= \sin x + 1$ x-intercept of 270°

ie $k = 2$